

● *Original Contribution*

INFLATION AND BI-AXIAL TENSILE TESTING OF HEALTHY PORCINE CAROTID ARTERIES

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Abstract—Knowledge of the intrinsic material properties of healthy and diseased arterial tissue components is of great importance in diagnostics. This study describes an *in vitro* comparison of 13 porcine carotid arteries using inflation testing combined with functional ultrasound and bi-axial tensile testing. The measured tissue behavior was described using both a linear, but geometrically non-linear, one-parameter (neo-Hookean) model and a two-parameter non-linear (Demiray) model. The shear modulus estimated using the linear model resulted in good agreement between the ultrasound and tensile testing methods, $G_{US} = 25 \pm 5.7$ kPa and $G_{TT} = 23 \pm 5.4$ kPa. No significant correspondence was observed for the non-linear model $a_{US} = 1.0 \pm 2.7$ kPa vs. $a_{TT} = 17 \pm 8.8$ kPa, $p \sim 0$; however, the exponential parameters were in correspondence ($b_{US} = 12 \pm 4.2$ vs. $b_{TT} = 10 \pm 1.7$, $p > 0.05$). Estimation of more complex models *in vivo* is cumbersome considering the sensitivity of the model parameters to small changes in measurement data and the absence of intraluminal pressure data, endorsing the use of a simple, linear model *in vivo*. (E-mail: rlopata@tue.nl) © 2016 World Federation for Ultrasound in Medicine & Biology.

Key Words: Ultrasound elastography, Carotid artery, Inflation testing, Bi-axial tensile testing.

INTRODUCTION

Methods of non-invasive diagnosis of arterial dysfunction are of great clinical interest (Ammirati et al. 2013; Czernuszewicz et al. 2015; Hansen et al. 2013; Magnoni et al. 2015). The current methods used to diagnose carotid atherosclerosis are based on geometric information, which does not provide any functional information on the vessel at risk and results in overtreatment of around 80% of the patient population (Medical Research Council 1998; Rothwell and Warlow 1999). Vascular strain imaging and elastography are non-invasive techniques, allowing the assessment of strains and mechanical properties of the arterial wall from ultrasound, respectively (Baldewsing et al. 2004; de Korte et al. 1998; Fromageau et al. 2008; Schmitt et al. 2007). These techniques, however, have been validated experimentally on simple phantoms with known properties (Cinthio et al. 2010; Larsson et al. 2015; Ribbers et al. 2007; Ryan and Foster 1997; Widman et al. 2015) and on biological vascular tissue

(Keyes et al. 2013; Lopata et al. 2014; van den Broek et al. 2011). Studies that compare functional ultrasound measurements with measurements obtained with another experimental method are scarce.

Several experimental methods used to acquire material properties of different vascular tissue components have been described in the literature. Tensile testing, either uni-axial or bi-axial, is most commonly applied. Both tests apply a stretch to the tissue, and the resulting force is measured. For vascular tissue, bi-axial tensile testing is preferred over uni-axial testing, because of the tissue's anisotropy and the loading of the tissue in all directions *in vivo*. Several studies have reported on uni- and bi-axial tensile testing of carotid arteries, either with or without atherosclerotic plaque (Choserot et al. 2005; Kural et al. 2012; Lawlor et al. 2011; Maher et al. 2009). Similar measurements were reported for coronary arteries (Lally et al. 2004) and healthy or aneurysmal aortic tissue (Duprey et al. 2010; Holzapfel and Ogden 2006; Nicosia et al. 2002; Okamoto et al. 2002). Inflation testing characterizes the arterial tissue in more physiologic circumstances *in vitro*, that is, less destructive circumstances, and was previously employed on several types of arteries (Agianniotis

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et al. 2012; Beattie *et al.* 1998; Boekhoven *et al.* 2014; Lopata *et al.* 2014; Schulze-Bauer *et al.* 2003; Vychytil *et al.* 2010).

For mechanical characterization of tissue, the change in both luminal diameter and wall thickness needs to be assessed. Moreover, the local radial displacements and strains in the wall can be of help in understanding the mechanical behavior of the tissue. Both can be measured with ultrasound (strain) imaging techniques, either *in vivo* or using the *in vitro* inflation method (Boekhoven *et al.* 2014; Hansen *et al.* 2013; van den Broek *et al.* 2011). From the radiofrequency (RF) data, local deformation fields are estimated and displayed as color overlays on the regular B-mode images, that is, strain imaging (Talhami *et al.* 1994). The combination of inflation testing and ultrasound strain imaging has been used in healthy and atherosclerotic tissue (Boekhoven *et al.* 2013; Ribbers *et al.* 2007) and in other types of arteries, mostly in a quasi-static fashion (Brusseau *et al.* 2001; de Korte *et al.* 1998; Lopata *et al.* 2014; Schaar *et al.* 2003).

One step beyond strain imaging is so-called elastography or elastometry. The aim in vascular elastography is to identify the local mechanical properties of vessels *in vivo*, using invasive or non-invasive ultrasound imaging (Baldewijs *et al.* 2004; Maurice *et al.* 2008). Elastometry is defined as the estimation of a global parameter set from ultrasound data (Bensamoun *et al.* 2008; Saito *et al.* 2004). Validation of vascular strain imaging- and vascular elastography-related techniques is based mainly on simulation and phantom studies. The main advantage of simulation data is the known ground truth of the motion and deformation; however, the contrast of the images and speckle patterns are unrealistic. The same disadvantages are valid in the phantom studies, which have the advantage of known material properties and geometry. For the validation of results from vascular elastometry, bi-axial tensile testing has proven to be a good candidate. The direction and magnitude of the forces acting on the vessel sample can be matched for both methods. Lopata *et al.* (2014) reported good agreement between the two methods when modeling the porcine aortic arterial wall as a neo-Hookean solid.

By choosing an appropriate material model, the intrinsic material properties of the tissue can be estimated, which requires information not only on deformation of the tissue, but also on the force causing it. Numerous material models are described in the literature. The arterial wall is mostly described as a linear or non-linear hyperelastic solid material, implying that the stress-strain relation is derived from a strain energy density function. The constitutive material models advance from one-parameter models (*e.g.*, neo-Hookean model), to two-parameter

models (*e.g.*, two-term Ogden, Fung, Demiray and Mooney-Rivlin models, Delfino *et al.* (1997), Fung *et al.* (1979), to non-linear, fiber-reinforced models such as those described by Driessen *et al.* (2005), Gasser *et al.* (2006) and Holzapfel and Gasser (2000). For translational purposes simple models are required, because the number of parameters to be obtained from clinical data is small. Moreover, the stress-strain relation of vascular tissue in the physiologic range is quite linear; hence the use of more simple models could be valid.

The goal of this study was to compare ultrasound elastographic measurements in carotid arteries obtained with inflation testing to those obtained with bi-axial tensile testing. In a previous study by our group, such a comparison was performed in relatively large vessels (aortas) using a simple, quasi-static approach (Lopata *et al.* 2014). In this study, 2-D RF-based displacement estimation was investigated for the much smaller carotid artery, which limits the performance because of the limitations in resolution. Inflation experiments were performed in a dedicated mock-circulation loop under more realistic, physiologic conditions with respect to pressurization, osmolarity and temperature. The complete mechanical behavior of 13 porcine carotids was assessed by pressurizing the wall for a full range of 0–120 mm Hg in a controllable and reproducible fashion. Moreover, dynamic pressurization was performed, which allowed testing for the more limited, physiologic pressure range, typically 80–120 mm Hg. Finally, mechanical behavior was described using a constitutive linear, one-parameter, material model (neo-Hookean) and a non-linear, two-parameter model (Demiray), for both the inflation and tensile testing data. The models were chosen to have a minimal number of parameters to allow accurate and robust estimation of the material parameters from this type of experimental data.

METHODS

Inflation testing

Sample preparation and data acquisition. Porcine carotid arteries were obtained from a local slaughterhouse ($n = 13$). The pigs were between 5 and 6 months of age and had a body weight in the range of 100–120 kg. The arterial segments were excised from the aortic arch to the bifurcation (common carotid artery), at the left side. The arteries were frozen at -20°C within 2 h of excision. For inflation testing, the carotid arteries were thawed, and excess material (mainly connective and adipose tissue) was removed from the total length of the segments and tested for leaks. Samples were discarded when leaks were found. The extracted samples had a stress-free length of about 30 mm and were selected to have a similar diameter (3.4 ± 0.5 mm). Cannulas were

inserted and suture fixed on both the proximal and distal ends of the artery. The artery was fixed in a mock circulation setup, described in [van den Broek et al. \(2011\)](#) and [Boekhoven et al. \(2013\)](#) (Fig. 1a). Next, the artery was pre-conditioned in both axial (z) and circumferential (y) directions, by stretching the artery in the axial direction five times from l_0 to its *in vivo* length, $\lambda_z = 1.6$, while being pressurized. The pre-stretch was determined from unpublished data, where axial force and pressure measurements were performed simultaneously during inflation testing, while the artery was being stretched in the axial direction. At a certain point the axial force will remain constant during dynamic pressurization, which equals the *in vivo* length of the artery. The arteries were perfused and surrounded with phosphate-buffered saline, at a temperature of 38.5°C. Each sample was continuously pressurized with a dynamic pulse (1 Hz), using a pump driven by a proportional pneumatic valve, with pressures ranging from 0 to 120 mm Hg at the rate of 1 Hz. This pressure range was chosen because the load-free situation is required for the model parameters. To verify the results obtained with the dynamic setup, 3 of 13 arteries were also subjected to a static inflation test. A water column was used to pressurize the samples from 0 to 120 mm Hg in steps of 10 mm Hg (~ 1 min

for each pressure step), similar to the study by [Lopata et al. \(2014\)](#).

Intraluminal pressure was recorded with a pressure wire (St. Jude Medical, St. Paul, MN, USA). The geometry and distension of the arteries were assessed using a MyLab70 ultrasound system (Esaote, Maastricht, Netherlands), equipped with a 2-D linear array transducer (LA-523), with a center frequency of 8 MHz, acquiring 38 RF lines/cm, including raw RF output. A longitudinal cross section of 34 mm was imaged for each artery during six pump cycles.

Data processing. The wall of the carotid artery was automatically segmented using the sustain attack filter. This filter is based on the generation of a reference signal to find the water wall boundaries. The threshold function is a decaying exponential function that is applied to the near and far walls individually ([Boekhoven et al. 2013](#); [Meinders et al. 2001](#)). The resulting segmentation was manually adjusted when necessary. Next, the original RF data were processed using a 2-D coarse-to-fine displacement estimation algorithm ([Lopata et al. 2009](#)). A two-step approach was used, for which the parameter settings are described in [Boekhoven et al. \(2014\)](#), resulting in displacement fields with a resolution of

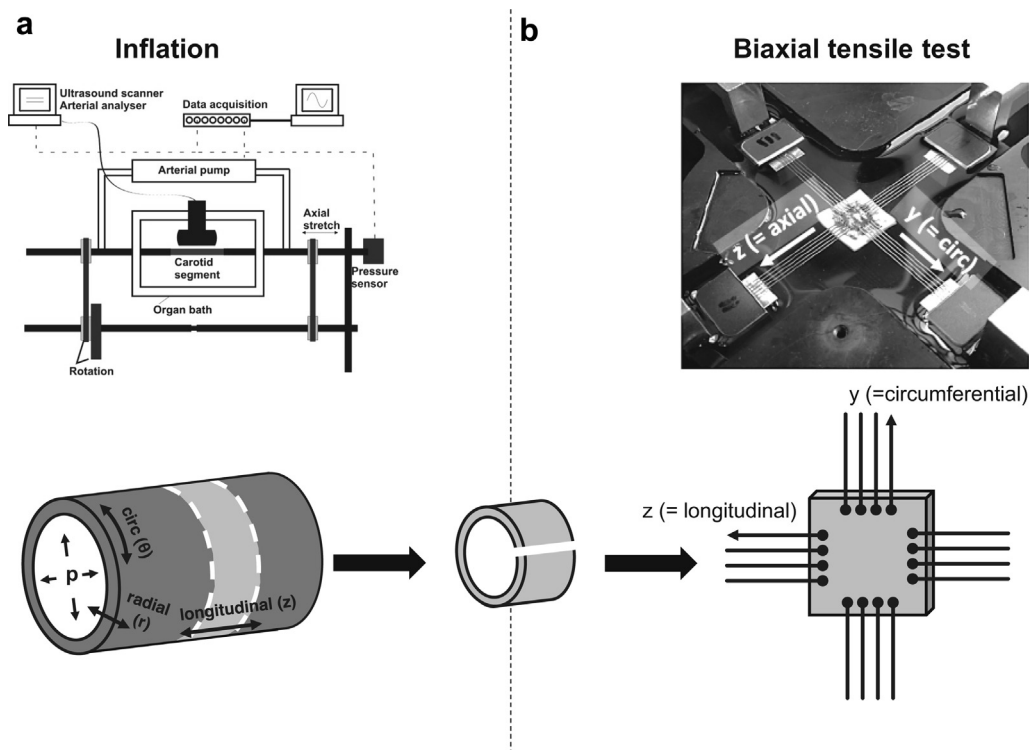


Fig. 1. Schematic representation of the two experimental methods used: (a) inflation testing and (b) bi-axial tensile testing. The coordinate systems used in this study are indicated in the graph, where x is the radial direction, y the circumferential direction and z the axial direction. Cylindrical coordinates were used for the inflation test (a); the tensile test was described in Cartesian coordinates (b).

0.3 × 0.7 mm, respectively axially and laterally. Next, the displacements were used to track the material points in the wall, as found by the sustain attack filter. From this, diameter, $d(t)$, and wall thickness, $h(t)$, were assessed over time, serving as input for the constitutive material models. All data analyses were based on the central 40 RF lines, equal to the center part (1 cm) of the artery, ensuring the lack of boundary effects. Furthermore, the analyses were carried out in the systolic phase, from the minimum (0 mm Hg) to the maximum (120 mm Hg) pressure.

Tensile testing

Sample preparation and data acquisition. After inflation testing, the porcine carotid arteries were cut in square samples of approximately 10 × 10 mm. Samples of tissue samples were obtained from the center of the artery in regions imaged during the inflation test. Tensile tests were performed in duplicate for each artery using two different samples (= 26 samples in total). The samples were marked with graphite particles, enabling deformation assessment via image analysis. Each sample was mounted into a bi-axial tensile testing device (CellScale). Four rakes, each consisting of five hooks, were attached to the sample, leading to a square testing area of 6.4 × 6.4 mm. The sample was excised from the carotid, ensuring that the two directions of tensile testing, that is, the directions on which the forces act on the tissue via the hooks, correspond to the circumferential and longitudinal directions of the vessel (Fig. 1b).

The samples were pre-conditioned by applying 10 cycles of a 1.1 stretch ratio in both directions. After pre-conditioning, a longitudinal pre-stretch of 1.6 was applied, corresponding to the pre-stretch in the inflation experiment. This stretch was kept constant during stretching in the circumferential direction (y-direction). The stretch in the y-direction was increased from 1.0 to 1.4 in steps of 0.05 at a stretch rate of 0.02/s. Between cycles, the tissue sample was unloaded for 5 s in the y-direction only, thereby matching the inflation experiment. This allowed the tissue and fibers to relax and return to their initial configuration. Images were taken by a digital photo camera at a frequency of 1 Hz. All tensile tests were performed at 38.5°C, matching the inflation experiments and body temperature.

Data processing. The images acquired during the tensile tests were analyzed using image registration software (LabJoy by CellScale). A square grid of 9 × 9 samples was placed in the center of the first image prior to circumferential stretching, corresponding to a grid of approximately 2 × 2 mm. Each grid point was tracked during the entire stretch cycle. The deformation fields

were computed from these data as described in [Geers et al. \(1996\)](#).

Mechanical characterization and validation

Neo-Hookean model. Two models were chosen to characterize the material properties of the vascular tissue. First, the neo-Hookean material model was used, thereby assuming non-linear stress strain behavior, isotropic material behavior and incompressibility of the tissue. The sole material parameter is the shear modulus G . The neo-Hookean model is valid for large strains, whereas Hooke's law is not. The model was fitted to both the inflation and tensile test data. The resulting material property, in this case the shear modulus, was compared between the two different experimental methods. The stress–strain relation for a neo-Hookean material is given by

$$\boldsymbol{\sigma} = -p_0\mathbf{I} + G(\mathbf{B} - \mathbf{I}) \quad (1)$$

with $\mathbf{B} = \mathbf{F}\mathbf{F}^T$, the left Cauchy–Green strain tensor, with $\mathbf{F}$ the deformation tensor, G the shear modulus and p_0 the undetermined hydrostatic stress. The diagonal components of the deformation tensor are given in cylindrical coordinates for the inflation test (the off-diagonal components are zero):

$$\lambda_r = \frac{R}{rk\lambda_z}, \lambda_\theta = \frac{rk}{R}, \lambda_z = \frac{l}{l_0} \quad (2)$$

Parameters r and R describe the radius in the deformed and stress-free configurations, respectively (as illustrated in [van der Horst et al. 2011](#)). The residual stress is taken into account by means of k , which is dependent of the opening angle α , where $k = 2\pi/(2\pi - \alpha)$. From eqns (1) and (2), the following stress–stretch relationship can be derived for the inflation test:

$$\sigma_{\theta\theta} - \sigma_{rr} = G_{US}(\lambda_\theta^2 - \lambda_r^2) \quad (3)$$

$\sigma_{\theta\theta}$ and σ_{rr} are the circumferential and radial engineering stresses, respectively. From the conservation of momentum, and assuming axisymmetry and neglecting edge effects, the following equation holds:

$$\frac{\delta\sigma_{rr}}{\delta r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0 \quad (4)$$

The classic solution to this differential equation by Lamé was employed (e.g., [El-Kurdi et al. 2008](#)) using the boundary conditions $\sigma_{rr}(r = r_o) = 0$ and $\sigma_{rr}(r = r_i) = -p$ to derive the Lamé stress components in radial and circumferential direction:

$$\sigma_{rr} = \frac{r_i^2 p}{r_o^2 - r_i^2} \left(1 - \frac{r_o^2}{r_i^2} \right), \text{ and } \sigma_{\theta\theta} = \frac{r_i^2 p}{r_o^2 - r_i^2} \left(1 + \frac{r_o^2}{r_i^2} \right) \quad (5)$$

with r_i and r_o the inner and outer radii, respectively, and p the intraluminal pressure. The mean wall stress difference can be calculated using integration, yielding

$$\overline{\sigma_{\theta\theta}} - \overline{\sigma_{rr}} = \frac{4 \cdot p \cdot r_o^2 \cdot r_i^2 \cdot \log(r_o/r_i)}{(r_o^2 - r_i^2)^2} \quad (6)$$

Similarly, for the tensile test, the following equation can be derived in Cartesian coordinates (see Lopata et al. [2014]),

$$\overline{\sigma_{yy}} = \frac{f_{\text{rakes}}}{A} = G_{\text{TT}} \left(\lambda_y^2 - \frac{1}{\lambda_y^2 \lambda_z^2} \right) \quad (7)$$

with the stress now calculated from the force measured, f_{rakes} , which is the force exerted during bi-axial testing in the circumferential direction, and A , the updated surface area on which the force was exerted. λ_y is the stretch in the circumferential direction, applied during the tensile testing experiment. The shear modulus, obtained by tensile testing, is referred to as G_{TT} , and λ_z was set to 1. The full derivation can be found in the Appendix of Lopata et al. (2014). Finally, the shear moduli G_{US} and G_{TT} were estimated from the experimental data using a least-squares estimation procedure implemented in MATLAB (The MathWorks, Natick, MA, USA). The allowed parameter range for G was set to 1 kPa–10 MPa for both inflation and tensile test data, and the initial guess was set to $G = 20$ kPa. To allow comparison of both methods, analyses were performed in the same stress range.

Demiray model. The Demiray (or Delfino) material model was chosen as an example of a two-parameter model, which has been used to describe the material behavior of cardiac tissue and carotids (Delfino 1996; Demiray 1972; Leach et al. 2010). Furthermore, this model exhibits stable behavior under dead-loading conditions (Ogden 2003). The strain energy density function is

$$W = \frac{a}{b} \left[\exp\left(\frac{b}{2}(I_1 - 3)\right) - 1 \right] \quad (8)$$

now with a the stiffness parameter, and b a dimensionless, non-linear parameter and I_1 the first invariant of B . Similar to the neo-Hookean model, the stress difference for the inflation test is

$$\sigma_{\theta\theta} - \sigma_{rr} = \frac{a}{2} (\lambda_\theta^2 - \lambda_r^2) e^{\frac{b}{2}(\lambda_r^2 + \lambda_\theta^2 + \lambda_z^2 - 3)} \quad (9)$$

And, for the tensile test,

$$\sigma_{yy} - \sigma_{xx} = \frac{a}{2} \left(\lambda_y^2 - \frac{1}{\lambda_y^2 \lambda_z^2} \right) e^{\frac{b}{2} \left(\frac{1}{\lambda_y^2 \lambda_z^2} + \lambda_y^2 + \lambda_z^2 - 3 \right)} \quad (10)$$

The two parameters a and b were both estimated using the same parameter fitting algorithm, again using the Lamé stress derivative as described in eqn (6). Based on a previous study (Delfino 1996), the allowed parameter ranges were set to: $a \in [0, 1 \times 10^5 \text{ kPa}]$, and $b \in [0, 50]$, and the initial guesses for a and b were 2.5 kPa and 3, respectively. Different starting parameters resulted in the same global minimum of the fitting procedure.

Data analysis

Data were analyzed for all experiments according to the flowchart in Figure 2.

Inflation testing versus tensile testing. The first goal of this study was to compare dynamic inflation testing with tensile testing. Arteries were tested over the full pressure range, 0–120 mm Hg, and corresponding stresses were applied during tensile testing. Material parameters were estimated from the data determined using both material models. A small phase difference existed between $p(t)$ and $d(t)$ for the dynamic data, which was corrected for by matching the maximum pressure, p , with the maximum diameter, d , and accompanying wall thickness, h . For comparison, the material parameters were estimated from the tensile testing data for matching stress ranges, where $\overline{\sigma_{rr}}$ was assumed to be $-p/2$. Residual stress was incorporated, where α was set to 74° (Vychytil et al. 2010).

Static versus dynamic. To compare quasi-static inflation with dynamic inflation testing, three arteries were inflated also in quasi-static fashion (Lopata et al. 2014). The material parameters were estimated for both data sets, again using both material models.

Incremental. The incremental shear moduli (G_{inc}) were estimated in both the inflation and the tensile testing data for the physiologic pressure range 80–120 mm Hg. The material behavior is expected to be linear in the physiologic pressure range, as was described by Debes and Fung (1995) and Sugawara et al. (2000).

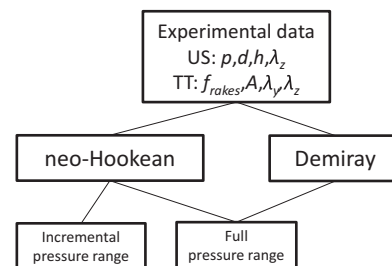


Fig. 2. Flowchart describing the experimental and numerical work in this study.

Statistical analysis

The parameter estimations of two experimental methods will be compared. This only includes a statistical comparison between dynamic inflation and the bi-axial tensile test, whereas static vs. dynamic only has $n = 3$. The experimental methods were performed on the same tissue; therefore, a paired Bland–Altman analysis was performed. The bias between the parameter estimations from both experimental methods was determined, and was tested using Student's t -test. When the bias is close to 0, correspondence between the methods is good.

Histology

Small rings with a thickness of approximately 1 mm were cut from the center of the vessels prior to tensile testing for histologic purposes. Vessel rings were fixed in 4% formaldehyde for 24 h and routinely processed for paraffin embedding using conventional techniques. To distinguish and quantify arterial structure, that is, the intima, media and adventitia, standard hematoxylin and eosin staining was performed. Histologic samples were visualized using the Zeiss Axio Observer Z1 inverted microscope outfitted with a Plan-Neofluar $5 \times /0.13$ objective (Carl Zeiss, Oberkochen, Germany).

RESULTS

Inflation testing versus tensile testing

In the stress–strain curves in Figure 3, the shaded areas indicate the variance (one standard deviation [SD]) in both Cauchy stress (Fig. 3a) and Finger strain (Fig. 3b) for all 13 carotid arteries, revealing the variation in responses of the different carotids. The stress–strain curves have a similar shape; however, because of the residual stress incorporated into the inflation experiment,

the onset of the strain is negative. This effect is not observed in the tensile data because the circumferential stretch (λ_z) was kept at 1. The Finger strain, at maximum Cauchy stress (39 ± 6.1 kPa), was $56 \pm 8.3\%$ for the inflation experiment and $57 \pm 8.5\%$ for the tensile test (referring to the mean $\pm$ SD in the remainder of this paper). Furthermore, the mean stress–strain curves of the inflation experiment (Fig. 4a) and the tensile testing experiment (Fig. 4b) were fitted with both material models; the incremental shear moduli were estimated as well, indicating that the Demiray model can describe the non-linear stress–strain relation properly. Mean parameter values are listed in Table 1.

Shear modulus (G) was determined, using the neo-Hookean material model, for each artery for both experiments, $G_{US} = 25 \pm 5.7$ kPa and $G_{TT} = 23 \pm 5.4$ kPa (Table 1). Bland–Altman analysis revealed a small bias ($G_{US} - G_{TT} = 2.4 \pm 5.1$ kPa). Moreover, all samples were found within the limits of agreement (Fig. 5). A paired t -test confirmed the finding of a fixed bias with $p < 0.05$.

No correspondence was observed between the estimated stiffness parameters (a) from the non-linear two-parameter model when comparing inflation and tensile testing; however, the non-linearity parameters (b) were in agreement. Parameter estimates were acquired at the local minimum of the cost function, resulting in good fits (mean r^2 of 0.99). However, the parameter estimates were spread widely (Fig. 6), with $a_{US} = 35 \pm 39$ Pa and $b_{US} = 12 \pm 2.5$ (median $\pm$ IQR). For tensile testing, $a_{TT} = 13 \pm 15$ kPa and $b_{US} = 10 \pm 2.2$ (see Table 1 for the mean values). No significant difference was found for the non-linear parameter b ($p < 0.05$). The Demiray parameters were determined for the mean stress–strain curves as well, represented by the red circles in Figure 6. For

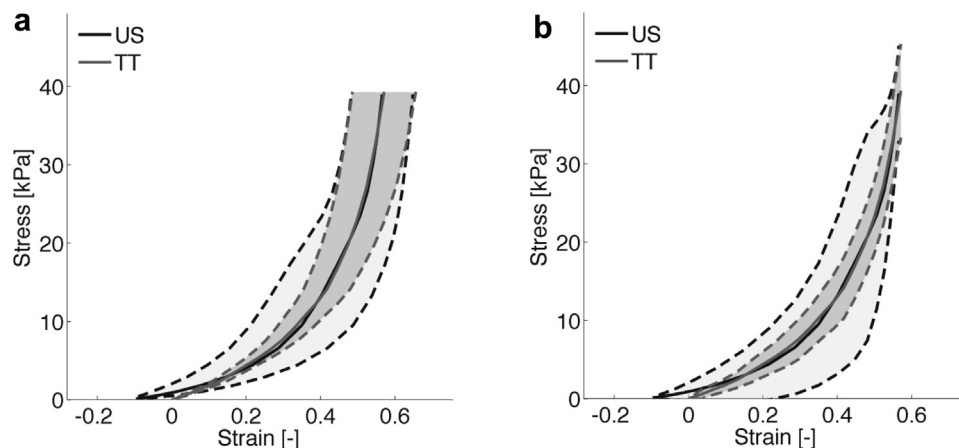


Fig. 3. Stress–strain curves for the inflation experiment (US) (black) and TT (gray). Shaded areas represent the variance within one standard deviation. (a) Variance in the Finger strain. (b) Variance in the Cauchy stress for all 13 carotid arteries. US = ultrasound; TT = tensile test.

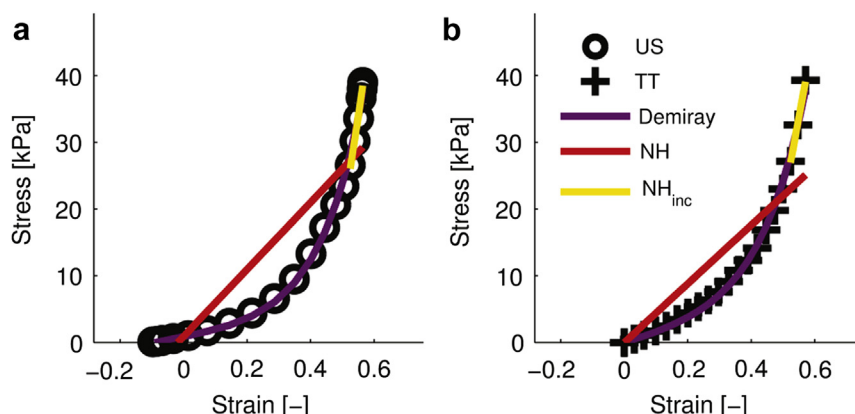


Fig. 4. (a) Mean stress–strain curve for the inflation experiment (US). (b) Mean stress–strain outcome of the bi-axial TT, with the fit for the neo-Hookean model in red for the full stress range and in yellow for the physiologic stress range. The fit for the Demiray model is in purple. US = ultrasound; TT = tensile test.

inflation testing, $a_{US} = 0.3$ kPa and $b_{US} = 8.7$, whereas for the tensile testing data, $a_{TT} = 16$ kPa and $b_{TT} = 9.4$. Hence, the neo-Hookean model reveals higher consistency and better agreement regarding material parameter estimation. Additionally, histology revealed no abnormalities in the structure of the carotid arteries.

Static versus dynamic

Pressure–diameter curves from four randomly chosen samples are provided in Figure 7a. In this figure, results from two subsequent pressure cycles are visualized, indicating the reproducibility of the pressure–diameter data. Three samples underwent both static and dynamic pressurization (Fig. 7b). In general, the pressure–diameter curve based on the static data had a different shape than that based on the dynamic data. The diameters had an acceptable deviation between the two experiments at the start (0 mm Hg) and end (120 mm Hg) points of <257 μm , which equals one wavelength of the ultrasound data acquired. This verifies that the starting point, that is, segmentation, and tracking were similar to those of the dynamic pressurization experiment. Moreover, G obtained from the dynamic data had a mean of 25 kPa (± 8.2 kPa), whereas for the static data the mean shear modulus was 21 kPa (± 4.4 kPa) ($n = 3$). The Demiray model was also fitted

to these data. For the dynamic data a range for a was found from 0.03 kPa to 3.2 kPa, and for b from 5.3 to 12. For the static data, consistency was higher, in the ranges 0.3 to 0.4 kPa for a and 7.9 to 8.8 for b .

Incremental

Incremental shear moduli were calculated for all arteries from a point where the Cauchy stress was equal to 25 kPa, which on average corresponded to 80 mm Hg in this study. The incremental shear modulus was 153 ± 71 kPa for the US data and 121 ± 25 kPa for the TT data. Bland–Altman analysis revealed a bias of 31 ± 64 kPa. A paired t -test confirmed the finding of a fixed bias with $p < 0.05$.

DISCUSSION

In this study, porcine carotid arteries were mechanically characterized by means of inflation testing and bi-axial tensile testing using both the neo-Hookean and Demiray material models. In the comparison of inflation experiments with tensile testing, little bias was observed between parameter estimates when the material was described as a neo-Hookean solid. All data revealed similar material behavior, resulting in comparable material properties, with a shear modulus around 24 kPa

Table 1. Material parameter estimates for the neo-Hookean model, as the Demiray strain energy density function for the inflation experiment, and the biaxial tensile test*

Measurement method	Neo-Hookean		Demiray	
	G (kPa)	G_{inc} (kPa)	a (kPa)	b
Ultrasound	25 ± 5.7 (3,425)	153 ± 71 (157)	1.0 ± 2.7 (0.3)	12 ± 4.2 (8.7)
Tensile test	23 ± 5.4 (3,622)	121 ± 26 (129)	17 ± 8.8 (16)	10 ± 1.7 (9.4)

* Estimates are displayed means $\pm$ standard deviations. Parameters estimated from the mean stress–strain curve are given in parentheses.

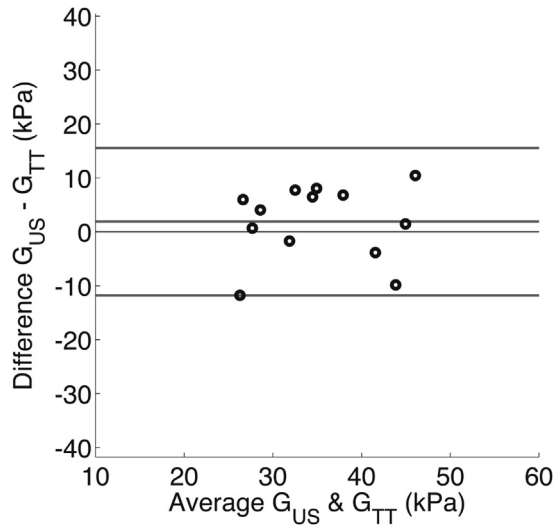


Fig. 5. Bland–Altman plot comparing the shear moduli estimated in the inflation experiment (G_{US}) and tensile test (G_{TT}).

(approximately agreement with Vychytil *et al.* [2010]), over the full pressure range at $\lambda_z = 1.6$. The stress–strain curves revealed a different behavior at the onset of inflation, caused by the residual strain, which resulted in compression at the onset of inflation.

Neo-Hookean model

As already mentioned, a similar study was previously performed on porcine aortas (Lopata *et al.* 2014); however, there are some methodologic differences from the current study. The carotid artery has a smaller diameter and thinner wall than the aorta. Hence, high resolution is needed and relatively small displacements need to be measured, approaching the physical limits of

the US system. Moreover, a dynamic pulse was applied to the artery. Although the pressure pulse had the same magnitude as the pressure applied in the static experiment, the pulsatile behavior may lead to different results because of visco-elastic behavior, which was actually observed in this study. The static experiment has the advantage of little out-of-plane motion and very controlled, more accurate pressurization. However, the small levels of visco-elastic behavior can be more dominant because of the longer time between subsequent pressure steps in the static approach. Finally, the carotid arteries were tested in phosphate-buffered saline and at body temperature, which was not the case in the study on aortas.

Considering the geometry of the carotids, the arteries were modeled as thick-walled cylinders, and residual stress was taken in account, resulting in a more reliable solution for parameter estimation. The opening angle was determined for several carotid rings; however, the opening angle measured ranged from 0° to 90° . Experimental determination of the opening angle is vulnerable to a large error, as discussed previously (Rachev and Greenwald 2003). The open-sector geometries do not resemble a perfect circle with uniform wall thickness. Van der Horst *et al.* (2011) reported the use of an optimization method to determine the opening angle of the coronary artery. As the axial force necessary for this optimization procedure was not given in that study, the value was set to 74° as reported in Vychytil *et al.* (2010).

The magnitudes of the stress–strain data found in this study are in correspondence with those reported in Cox (1975) and Silver *et al.* (2003). Studies comparing inflation testing and bi-axial tensile testing are, however, rather scarce. In a study by Keyes *et al.* (2013), there was a significant difference in estimating the circumferential mean tangential modulus between *in vitro* inflation testing and bi-axial tensile testing on porcine coronary arteries. They determined the outer diameter change optically. Also, US was used to determine diameter changes in the coronaries while the artery was still embedded in the heart. These data revealed good correspondence with the bi-axial tensile test. Other similar studies determined mainly the Young's modulus and found good agreement between inflation experiments and bi-axial tensile testing (Dutta *et al.* 2013; Kim *et al.* 2008). Although the Young's modulus is also a linear elastic parameter, it is valid only for small strains, whereas the neo-Hookean model can also be used to describe large deformations such as those found in this study.

Some spread was observed within the different testing methods. The shear modulus had a standard deviation of approximately 6 kPa for both experimental methods. This could be explained by differences in

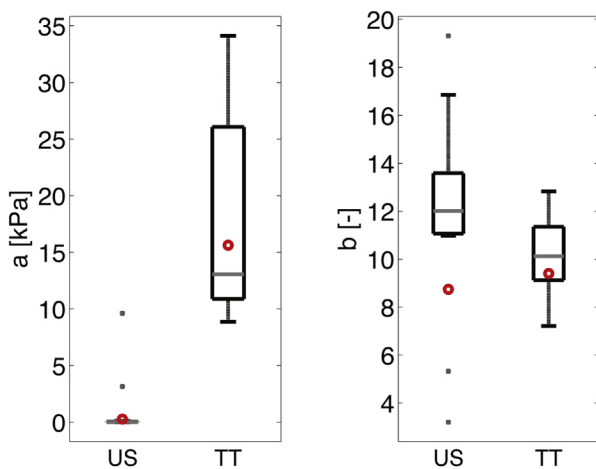


Fig. 6. Boxplot representation of Demiray parameter estimates (a and b) in the full pressure range, for inflation testing (US) and TT. The red circles represent the parameter estimates from the mean stress–strain curve. US = ultrasound; TT = tensile test.

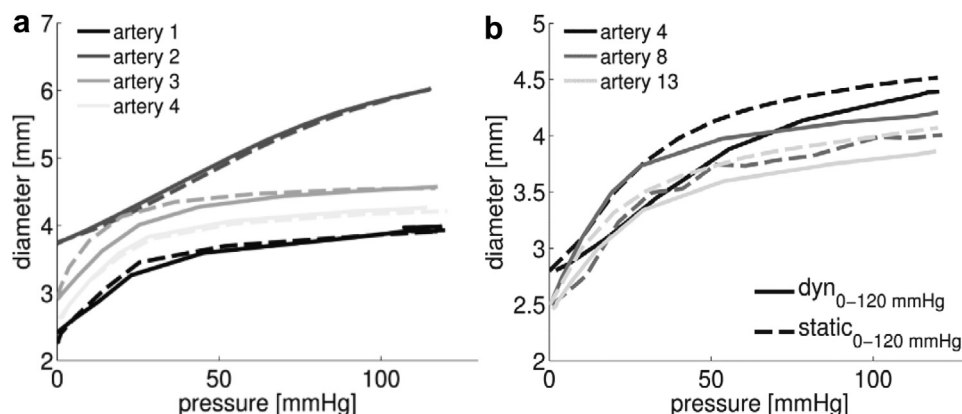


Fig. 7. (a) Reproducibility for four different arteries (Nos. 1–4) in the pressure–diameter curves for two different pressure cycles. (b) Pressure–diameter curves of arteries 4, 8 and 13 for the static and dynamic inflation experiments.

morphology, measurement errors and biological variability. Results from histology revealed that the intima–media complex was still intact and that the adventitial layer was still present, indicating that the mechanical behavior of the complete vessel wall was tested here. Moreover, some loose connective tissue was not completely removed for all vessels. This tissue was assumed to have no significant role in the mechanical behavior of the vessel wall.

Segmentation of the vessel wall was carried out automatically, with the exception of some manual adjustments in case of obvious errors, possibly caused by reflections of said loose connective tissue. The resulting overestimation of vessel wall thickness will result in underestimation of the shear modulus, which possibly was the case in one sample (No. 10). Another source of error could be the longitudinal positioning of the arterial segment within the pig. The porcine common carotid changes from a muscular artery at the distal end to an elastic vessel near the aortic arch. Despite the strict protocol, differences in anatomy between pigs could have resulted from the acquisition of samples at different locations along the length of the artery. However, from the histology, no clear relation was found between the amount of media and shear modulus.

With respect to the differences between static and dynamic inflation testing of the arteries, a stiffer response (higher G) was observed for the dynamic condition, which was confirmed in a study by Lichtenstein et al. (1998). The different responses of the diameter to the applied pressure, either static or dynamic, are due to the visco-elastic behavior of the arterial wall. The mechanical response of a visco-elastic material depends both on the force applied (elastic) and on the time it acts (viscous). We found a small bias between dynamic inflation and the bi-axial test, whereas the bias between static inflation and bi-axial testing was smaller, in agreement

with Lopata et al. (2014). This inconsistency between studies could be attributed to the stretching of the inner layer caused by flattening the tissue at the onset of the bi-axial tensile test, in the current study, which was less relevant for the relatively thin-walled aorta.

Demiray model

In the more complex, non-linear, two-parameter Demiray model, a larger spread was observed in the parameter estimates, and no clear agreement was found between the two experimental methods for the stiffness parameter a . The mean stiffness parameter a was approximately 17 times higher for the tensile testing data. The negative onset of inflation testing caused by pre-stretching explains the difference in a . The exponential coefficient b revealed more similarity between the two experiments, which is in correspondence with the shapes of both stress–strain curves (Fig. 3). The large spread in parameter estimates could be caused by the non-uniqueness problems that can occur when using a least-squares optimization procedure, which is due to the associated sensitivity to small changes in the data (Fung 1990; Holzapfel and Gasser 2000). Furthermore, this large spread in the parameter estimates is a result of relatively small deviations in the raw data. In clinical studies, the spread in data is expected to be larger; therefore, translation to clinical studies using this model would not be straightforward.

Moreover, stress is not equally distributed over the vessel wall layers during inflation testing, whereas it is approximately equally distributed for tensile testing. This can result in a smaller contribution by the adventitia and greater contribution by the intimal layer. Delfino (1996) performed inflation experiments on aged (30–42 y) human carotid arteries from autopsy, using a thick-wall approach. Parameters were estimated to be 44 kPa for a and 16 for b . The lower stiffness parameter

found in this study can be explained by the use of porcine material obtained from young animals. This is corroborated by the fact that at 120 mm Hg, Delfino (1996) obtained a circumferential stretch of 1.2, whereas in this study, the average circumferential stretch was 1.6. The mutual dependence of the two parameters also influences b ; however, it must be noted that Delfino (1996) applied an axial pre-stretch of 1.1 (obtained from *in situ* measurements).

In a study of Keyes *et al.* (2013), a model by Fung was fitted to similar data obtained in coronary arteries. No match in the stiffness parameter was found when comparing inflation and bi-axial testing, similar to this study. The lower stiffness in the bi-axial testing data in that study was also attributed to the uncrimping of collagen fibers when flattening the tissue, which was observed in microstructural analysis. Another model commonly used to describe the arterial wall is that of Holzapfel and Gasser (2000), which also takes into account the anisotropy of the vessel wall. Two stiffness parameters, linear and non-linear, a dimensionless parameter, and a fiber angle need to be estimated, vastly increasing the complexity of the model. This model has been applied to coronary data as described by van der Horst *et al.* (2011) and was also considered for this study. However, a limitation of the current *in vitro* study is the lack of axial force measurements, which were used to enhance the accuracy of the non-linear model fit, while reducing the spread in parameter estimates. However, it would never be possible to obtain the axial force data *in vivo*.

Future work, recommendations

Future work includes translation toward the clinic through characterization of the vessel using dynamic data obtained in the range 80–120 mm Hg only. An important issue here is accurate segmentation of both the lumen and wall thickness. The most appropriate path would be to assume the wall thickness to be equal to the intima-media thickness, which can already be determined automatically in the clinic using ultrasound (Beaussier *et al.* 2008; Bianchini *et al.* 2010). However, *in vivo* conditions are less controlled, including the pressure and resulting levels of strain and the amount of out-of-plane motion present. Moreover, the presence of active elastic properties in the *in vivo* situation, as opposed to the passive mechanical properties determined in this study, will most likely result in a different outcome. Furthermore, the acquisition of pressure–time data, which are required for the modeling equations, is not trivial *in vivo*. In addition, for the non-linear model, the situation at zero pressure needs to be known and will likely introduce much uncertainty in the model parameter estimates. Hence, estimation of an incremental modulus is probably more feasible, especially when considering the extent to which data can be obtained *in vivo*.

CONCLUSIONS

In this study, a vascular elastometry method was compared with tensile testing in porcine carotid arteries. For the Demiray model, little agreement was found between the two methods because of the difference in boundary conditions between the experiments. The neo-Hookean model however, was not sensitive to this difference, and agreement between the moduli obtained with both experimental methods was good. Hence, it is possible to estimate material parameters using a simple mechanical model, as was validated with tensile testing.

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REFERENCES

- Agianniotis A, Rachev A, Stergiopoulos N. Active axial stress in mouse aorta. *J Biomech* 2012;45:1924–1927.
- Ammirati E, Magnoni M, Camici PG. Need for new non-invasive imaging strategies to identify high-risk asymptomatic patients with carotid stenosis. *Int J Cardiol* 2013;168:4342–4343.
- Baldewing RA, de Korte CL, Schaar JA, Mastik F, van der Steen AFW. A finite element model for performing intravascular ultrasound elastography of human atherosclerotic coronary arteries. *Ultrasound Med Biol* 2004;30:803–813.
- Beattie D, Xu C, Vito R, Glagov S, Whang M. Mechanical analysis of heterogeneous, atherosclerotic human aorta. *J Biomech Eng* 1998; 120:602–607.
- Beaussier H, Masson I, Collin C, Bozec E, Laloux B, Calvet D, Zidi M, Boutouyrie P, Laurent S. Carotid plaque, arterial stiffness gradient, and remodeling in hypertension. *Hypertension* 2008;52:729–736.
- Bensamoun SF, Wang L, Robert L, Charleux F, Latrive JP, Ho Ba Tho MC. Measurement of liver stiffness with two imaging techniques: Magnetic resonance elastography and ultrasound elastometry. *J Magn Reson Imaging* 2008;28:1287–1292.
- Bianchini E, Bozec E, Gemignani V, Fata F, Giannarelli C, Ghiadoni L, Demi M, Boutouyrie P, Laurent S. Assessment of carotid stiffness and intima-media thickness from ultrasound data comparison between two methods. *J Ultrasound Med* 2010;29:1169–1175.
- Boekhoven RW, Lopata RGP, van Sambeek MR, van de Vosse FN, Rutten MCM. A novel experimental approach for three-dimensional geometry assessment of calcified human stenotic arteries *in vitro*. *Ultrasound Med Biol* 2013;39:1875–1886.
- Boekhoven RW, Rutten MCM, van Sambeek MR, van de Vosse FN, Lopata RGP. Echo-computed tomography strain imaging of healthy and diseased carotid specimens. *Ultrasound Med Biol* 2014;40: 1329–1342.
- Brusseau E, Fromageau J, Finet G, Delacharte P, Vray D. Axial strain imaging of intravascular data: Results on polyvinyl alcohol cryogel phantoms and carotid artery. *Ultrasound Med Biol* 2001;27: 1631–1642.
- Choserot C, Pry E, Goebel JC, Dumas D, Didelon J, Stoltz JF, Blondel WCPM. Experimental comparison between autofluorescence spectra of constrained fresh and cryopreserved arteries. *Clin Hemorheol Microcirc* 2005;33:235–242.
- Cinthio M, Jansson T, Eriksson A, Ahlgren ÅR, Persson HW, Lindström K. Evaluation of an algorithm for arterial lumen diameter measurements by means of ultrasound. *Med Biol Eng Computing* 2010;48:1133–1140.
- Cox RH. Arterial wall mechanics and composition and the effects of smooth muscle activation. *Am J Physiol* 1975;229:807–812.
- Czernuszewicz TJ, Homeister JW, Caughey MC, Farber MA, Fulton JJ, Ford PF, Marston WA, Vallabhaneni R, Nichols TC, Gallippi CM.

- Non-invasive *in vivo* characterization of human carotid plaques with acoustic radiation force impulse ultrasound: Comparison with histology after endarterectomy. *Ultrasound Med Biol* 2015;41:685–697.
- De Korte CL, Cespedes EI, van der Steen AF, Pasterkamp G, Bom N. Intravascular ultrasound elastography: Assessment and imaging of elastic properties of diseased arteries and vulnerable plaque. *Eur J Ultrasound* 1998;7:219–224.
- Debes JC, Fung Y. Biaxial mechanics of excised canine pulmonary arteries. *Am J Physiol Heart Circ Physiol* 1995;269:H433–H442.
- Delfino A. Analysis of stress field in a model of the human carotid bifurcation. Ph.D. thesis, École Polytechnique Fédérale de Lausanne, 1996.
- Delfino A, Stergiopoulos N, Moore JE, Meister JJ. Residual strain effects on the stress field in a thick wall finite element model of the human carotid bifurcation. *J Biomech* 1997;30:777–786.
- Demiray H. A note on the elasticity of soft biological tissues. *J Biomech* 1972;5:309–311.
- Driessen NJB, Bouten CVC, Baaijens FPT. A structural constitutive model for collagenous cardiovascular tissues incorporating the angular fiber distribution. *J Biomech Eng* 2005;127:494–503.
- Duprey A, Khanafar K, Schlicht M, Avril S, Williams D, Berguer R. *In vitro* characterisation of physiological and maximum elastic modulus of ascending thoracic aortic aneurysms using uniaxial tensile testing. *Eur J Vasc Endovasc Surg* 2010;39:700–707.
- Dutta D, Lee KW, Allen RA, Wang Y, Brigham JC, Kim K. Non-invasive assessment of elastic modulus of arterial constructs during cell culture using ultrasound elasticity imaging. *Ultrasound Med Biol* 2013;39:2103–2115.
- El-Kurdi MS, Viperman JS, Vorp DA. Control of circumferential wall stress and luminal shear stress within intact vascular segments perfused *ex vivo*. *J Biomech Eng* 2008;130:051003.
- Fromageau J, Lerouge S, Maurice RL, Soulez G, Cloutier G. Noninvasive vascular ultrasound elastography applied to the characterization of experimental aneurysms and follow-up after endovascular repair. *Phys Med Biol* 2008;53:6475.
- Fung YC. *Biomechanics: Motion, flow, stress, and growth*, vol 990. New York: Springer-Verlag; 1990.
- Fung YC, Fronek K, Patitucci P. Pseudoelectricity of arteries and the choice of its mathematical expression. *Am J Physiol Heart Circ Physiol* 1979;237:H620–H631.
- Gasser TC, Ogden RW, Holzapfel GA. Hyperelastic modelling of arterial layers with distributed collagen fibre orientations. *J R Soc Interface* 2006;3:15–35.
- Geers M, De Borst R, Brekelmans W. Computing strain fields from discrete displacement fields in 2-D-solids. *Int J Solids Struct* 1996;33:4293–4307.
- Hansen HHG, Richards MS, Doyley MM, de Korte CL. Noninvasive vascular displacement estimation for relative elastic modulus reconstruction in transversal imaging planes. *Sensors (Basel)* 2013;13:3341–3357.
- Holzapfel G, Gasser T. A new constitutive framework for arterial wall mechanics and a comparative study of material models. *J Elast Phys Sci Solids* 2000;61:1–48.
- Holzapfel G, Ogden R. (eds). *Mechanics of biological tissue*. New York/Berlin: Springer; 2006.
- Keyes JT, Lockwood DR, Utzinger U, Montilla LG, Witte RS, Vande Geest JP. Comparisons of planar and tubular biaxial tensile testing protocols of the same porcine coronary arteries. *Ann Biomed Eng* 2013;41:1579–1591.
- Kim K, Jeong CG, Hollister SJ. Non-invasive monitoring of tissue scaffold degradation using ultrasound elasticity imaging. *Acta Biomater* 2008;4:783–790.
- Kural MH, Cai M, Tang D, Gwyther T, Zheng J, Billiar KL. Planar biaxial characterization of diseased human coronary and carotid arteries for computational modeling. *J Biomech* 2012;45:790–798.
- Lally C, Reid AJ, Prendergast PJ. Elastic behavior of porcine coronary artery tissue under uniaxial and equibiaxial tension. *Ann Biomed Eng* 2004;32:1355–1364.
- Larsson M, Heyde B, Kremer F, Brodin LK, D'hooge J. Ultrasound speckle tracking for radial, longitudinal and circumferential strain estimation of the carotid artery: An *in vitro* validation via sonomicrometry using clinical and high-frequency ultrasound. *Ultrasonics* 2015;56:399–408.
- Lawlor MG, O'Donnell MR, O'Connell BM, Walsh MT. Experimental determination of circumferential properties of fresh carotid artery plaques. *J Biomech* 2011;44:1709–1715.
- Leach JR, Rayz VL, Soares B, Wintermark M, Mofrad MRK, Saloner D. Carotid atheroma rupture observed *in vivo* and FSI-predicted stress distribution based on pre-rupture imaging. *Ann Biomed Eng* 2010;38:2748–2765.
- Lichtenstein O, Safar ME, Mathieu E, Poitevin P, Levy BI. Static and dynamic mechanical properties of the carotid artery from normotensive and hypertensive rats. *Hypertension* 1998;32:346–350.
- Lopata RGP, Nillesen MM, Hansen HHG, Gerrits IH, Thijssen JM, de Korte CL. Performance evaluation of methods for two-dimensional displacement and strain estimation using ultrasound radio frequency data. *Ultrasound Med Biol* 2009;35:796–812.
- Lopata RGP, Peters MFJ, Nijs J, Oomens CWJ, Ruten MCM, van de Vosse FN. Vascular elastography: A validation study. *Ultrasound Med Biol* 2014;40:1882–1895.
- Magnoni M, Ammirati E, Camici PG. Non-invasive molecular imaging of vulnerable atherosclerotic plaques. *J Cardiol* 2015;65:261–269.
- Maher E, Creane A, Sultan S, Hynes N, Lally C, Kelly DJ. Tensile and compressive properties of fresh human carotid atherosclerotic plaques. *J Biomech* 2009;42:2760–2767.
- Maurice RL, Soulez G, Giroux MF, Cloutier G. Noninvasive vascular elastography for carotid artery characterization on subjects without previous history of atherosclerosis. *Med Phys* 2008;35:3436–3443.
- Medical Research Council. Randomised trial of endarterectomy for recently symptomatic carotid stenosis: final results of the MRC European Carotid Surgery Trial (ECST). *Lancet* 1998;351:1379–1387.
- Meinders JM, Brands PJ, Willigers JM, Kornet L, Hoeks AP. Assessment of the spatial homogeneity of artery dimension parameters with high frame rate 2-D B-mode. *Ultrasound Med Biol* 2001;27:785–794.
- Nicosia MA, Kasalko JS, Cochran RP, Einstein DR, Kunzelman KS. Biaxial mechanical properties of porcine ascending aortic wall tissue. *J Heart Valve Dis* 2002;11:680–686. discussion 686–687.
- Ogden R. *Biomechanics of soft tissue in cardiovascular systems*. New York: Springer; 2003.
- Okamoto RJ, Wagenseil JE, DeLong WR, Peterson SJ, Kouchoukos NT, Sundt TM III. Mechanical properties of dilated human ascending aorta. *Ann Biomed Eng* 2002;30:624–635.
- Rachev A, Greenwald S. Residual strains in conduit arteries. *J Biomech* 2003;36:661–670.
- Ribbers H, Lopata RGP, Holewijn S, Pasterkamp G, Blankensteijn JD, de Korte CL. Noninvasive two-dimensional strain imaging of arteries: Validation in phantoms and preliminary experience in carotid arteries *in vivo*. *Ultrasound Med Biol* 2007;33:530–540.
- Rothwell PM, Warlow CP. Prediction of benefit from carotid endarterectomy in individual patients: A risk-modelling study. European Carotid Surgery Trialists' Collaborative Group. *Lancet* 1999;353:2105–2110.
- Ryan L, Foster F. Ultrasonic measurement of differential displacement and strain in a vascular model. *Ultrason Imaging* 1997;19:19–38.
- Saito H, Tada S, Nakamoto N, Kitamura K, Horikawa H, Kurita S, Saito Y, Iwai H, Ishii H. Efficacy of non-invasive elastometry on staging of hepatic fibrosis. *Hepatol Res* 2004;29:97–103.
- Schaar JA, De Korte CL, Mastik F, Strijder C, Pasterkamp G, Boersma E, Serruys PW, Van Der Steen AFW. Characterizing vulnerable plaque features with intravascular elastography. *Circulation* 2003;108:2636–2641.
- Schmitt C, Soulez G, Maurice RL, Giroux MF, Cloutier G. Noninvasive vascular elastography: Toward a complementary characterization tool of atherosclerosis in carotid arteries. *Ultrasound Med Biol* 2007;33:1841–1858.
- Schulze-Bauer CAJ, Mörth C, Holzapfel GA. Passive biaxial mechanical response of aged human iliac arteries. *J Biomech Eng* 2003;125:395–406.

- Silver FH, Snowhill PB, Foran DJ. Mechanical behavior of vessel wall: A comparative study of aorta, vena cava, and carotid artery. *Ann Biomed Eng* 2003;31:793–803.
- Sugawara M, Niki K, Furuhashi H, Ohnishi S, Suzuki S. Relationship between the pressure and diameter of the carotid artery in humans. *Heart Vessels* 2000;15:49–51.
- Talhami H, Wilson L, Neale M. Spectral tissue strain: A new technique for imaging tissue strain using intravascular ultrasound. *Ultrasound Med Biol* 1994;20:759–772.
- Van den Broek CN, van der Horst A, Rutten MCM, van de Vosse FN. A generic constitutive model for the passive porcine coronary artery. *Biomech Model Mechanobiol* 2011;10:249–258.
- Van der Horst A, van den Broek CN, van de Vosse FN, Rutten MCM. The fiber orientation in the coronary arterial wall at physiological loading evaluated with a two-fiber constitutive model. *Biomech Model Mechanobiol* 2011;11:533–542.
- Vychytil J, Moravec F, Kochová P, Kuncová J, Švíglerová J. Modeling of the mechanical behaviour of porcine carotid artery undergoing inflation-deflation test. *Appl Comput Mech* 2010;4: 251–262.
- Widman E, Caidahl K, Heyde B, D'hooge J, Larsson M. Ultrasound speckle tracking strain estimation of *in vivo* carotid artery plaque with *in vitro* sonomicrometry validation. *Ultrasound Med Biol* 2015;41:77–88.